

7. Symmetries and classifications of free-fermion TI/TSC

7.1. Symmetries in quantum systems

Goal: Difference and relation of

① Symmetries of operators acting on Hilbert space

② Symmetries of physical states.

- Hilbert space $\neq$ physical states space

$\mathcal{H} \cong \mathbb{C}^N$, two states $|\psi\rangle$ and $|\psi'\rangle$ describes the same physical states iff $|\psi\rangle = z|\psi'\rangle$ for $0 \neq z \in \mathbb{C}$.

physical states space $P\mathcal{H} = \mathbb{CP}^{N-1} \cong (\mathcal{H} - \{0\}) / \mathbb{C}^\times$.

Ex. spin $\frac{1}{2}$, $\mathcal{H} = \mathbb{C}^2 = \{a|\uparrow\rangle + b|\downarrow\rangle \mid a, b \in \mathbb{C}\} = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{C}^2 \right\}$

$$\begin{aligned} \psi \begin{pmatrix} a \\ b \end{pmatrix} &\begin{cases} \textcircled{1} a \neq 0: \begin{pmatrix} a \\ b \end{pmatrix} \sim \frac{1}{\sqrt{|a|^2 + |b|^2}} \begin{pmatrix} |a| \\ |a| \cdot b/a \end{pmatrix} \sim \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix} \\ \textcircled{2} a = 0: \begin{pmatrix} 0 \\ b \end{pmatrix} \sim \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases} \end{aligned}$$

$$\Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} \sim \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix} \text{ with } \begin{cases} 0 \leq \theta \leq \pi \\ 0 \leq \phi < 2\pi \end{cases}$$

$$\Rightarrow P\mathcal{H} \cong \mathbb{CP}^1 \cong S^2 \rightarrow \text{Bloch sphere.}$$

$$\dim_{\mathbb{R}} \mathcal{H} = 4, \quad \dim_{\mathbb{R}} P\mathcal{H} = 2$$

normalization -1
U(1) phase -1

- Symmetries of Hilbert space $\neq$ Symmetries of physical states.

$\text{Aut}_{\text{qtm}}(P\mathcal{H})$: set of automorphisms of quantum systems.

transformation should preserve probability

$$|\langle \psi | \psi \rangle|^2 = |\langle U\psi | U\psi \rangle|^2$$

$\text{Aut}_{\mathbb{R}}(\mathcal{H})$: set of unitary and antiunitary transformations on $\mathcal{H}$.

$$U: |\psi\rangle \mapsto U|\psi\rangle$$

U is $\mathbb{R}$ -linear

$$\begin{cases} \text{unitary} & : & U(a|\psi\rangle + b|\phi\rangle) = a U|\psi\rangle + b U|\phi\rangle \\ \text{anti-unitary} & : & U(a|\psi\rangle + b|\phi\rangle) = a^* U|\psi\rangle + b^* U|\phi\rangle \end{cases}$$

Ex. $\text{spin } \frac{1}{2}$. $\mathcal{P}\mathcal{H} = S^2$, $\mathcal{H} = \mathbb{C}^2$

$$\text{Aut}_{\text{qtm}}(\mathcal{P}\mathcal{H}) = \text{Aut}_{\text{qtm}}(S^2) \cong O(3) \cong \mathbb{Z}_2 \times SO(3)$$

$$\begin{aligned} \text{Aut}_{\mathbb{R}}(\mathcal{H}) &= \text{Aut}_{\mathbb{R}}(\mathbb{C}^2) = \{\text{unitary/antiunitary transf. acting on } \mathbb{C}^2\} \\ &= U(2) \oplus U(2) = \mathbb{Z}_2 \times U(2) \end{aligned}$$

Relation:

$$0 \rightarrow U(1) \rightarrow \text{Aut}_{\mathbb{R}}(\mathcal{H}) \xrightarrow{\pi} \text{Aut}_{\text{qtm}}(\mathcal{P}\mathcal{H}) \rightarrow 0$$

Wigner thm: Every quantum automorphism in $\text{Aut}_{\text{qtm}}(\mathcal{P}\mathcal{H})$ is induced by a unitary or antiunitary operator in $\text{Aut}_{\mathbb{R}}(\mathcal{H})$ on Hilbert space $\mathcal{H}$.

$$1 \rightarrow U(\mathcal{H}) \rightarrow \text{Aut}_{\mathbb{R}}(\mathcal{H}) \xrightarrow{\phi} \mathbb{Z}_2 \rightarrow 0$$

$$\phi(s) = \begin{cases} +1, & \text{if } s \text{ is unitary} \\ -1, & \text{if } s \text{ is antiunitary} \end{cases}$$

- Physical symm and twisted symm.

$\text{Aut}_{\text{qtm}}(\mathcal{P}\mathcal{H})$: all symmetries of a q system.

Add Hamiltonian $\hat{H}$: smaller symmetry G .

$$\rho: G \rightarrow \text{Aut}_{\text{qtm}}(\mathcal{P}\mathcal{H})$$

twisted extension G^{tw} of G :

$$1 \rightarrow U(1) \rightarrow G^{\text{tw}} \rightarrow G \rightarrow 1$$

$$\begin{array}{ccccc} & & \parallel & \downarrow & \downarrow \\ & & & & \end{array}$$

$$1 \rightarrow U(1) \rightarrow \text{Aut}_{\mathbb{R}}(\mathcal{H}) \rightarrow \text{Aut}_{\text{qtm}}(\mathcal{P}\mathcal{H}) \rightarrow 1$$

$\left\{ \begin{array}{l} G: \text{physical symmetry acting on physical states} \\ G^{tw}: \text{virtual symmetry acting on Hilbert space.} \end{array} \right.$

Ex. $G = \mathbb{Z}_2^T = \{1, T\}$ is time reversal symmetry.

$$\begin{cases} \phi(1) = 1 \\ \phi(T) = -1 \end{cases}$$

$$1 \rightarrow U(1) \rightarrow G^{tw} \rightarrow \mathbb{Z}_2^T \rightarrow 1$$

G^{tw} is classified by $H^2(\mathbb{Z}_2^T, U(1)) = \mathbb{Z}_2 \ni \omega_2$

$$\begin{cases} (1) G^{tw} \cong \{zT \mid zT = Tz^{-1}, z \in U(1), T^2 = 1\} \cong U(1) \rtimes \mathbb{Z}_2^T \\ (2) G^{tw} \cong \{zT \mid zT = Tz^{-1}, z \in U(1), T^2 = -1\} \cong U(1) \rtimes_{\omega_2} \mathbb{Z}_2^T \end{cases}$$

7.2. 10-fold way of TI/TSC.

- unitary symmetries are NOT important in the classification of TI/TSC.

$$\hat{H} = \sum_{A,B} \hat{\psi}_A^\dagger \mathcal{H}_{A,B} \hat{\psi}_B \quad \mathcal{H}^T \sim \mathcal{H}^*$$

$$\{\psi_A, \psi_B^\dagger\} = \delta_{AB}, \quad \{\psi_A, \psi_B\} = \{\psi_A^\dagger, \psi_B^\dagger\} = 0.$$

If we have a unitary symmetry:

$$\begin{cases} \hat{U} \hat{\psi}_A \hat{U}^\dagger = \sum_B U_{A,B}^+ \hat{\psi}_B \\ \hat{U} \hat{\psi}_A^\dagger \hat{U}^\dagger = \sum_B \hat{\psi}_B^\dagger U_{B,A} \end{cases}$$

U is unitary matrix.

$$\hat{U} \hat{H} \hat{U}^{-1} = \hat{H} \iff U \mathcal{H} U^\dagger = \mathcal{H}$$

We can block-diagonalize $\mathcal{H}$:

$$\mathcal{H} = \bigoplus_{\lambda} \mathcal{H}_{\lambda}$$

For each λ , we have several copies of a fixed irrep. of G .

$$V_{\lambda} = V_{\lambda}^{(H)} \otimes V_{\lambda}^{(G)}$$

Hamiltonian
acting on

Symmetry
acting on

Within $V_\lambda^{(H)}$, there is no constraints for $\mathcal{H}$.

Ex. $G = \text{SO}(3)$. $V = \mathbb{C}^6 = \underbrace{\mathbb{C}_\lambda^3}_{\lambda = \text{spin}-1} \oplus \underbrace{\mathbb{C}_\lambda^3}_{\lambda = \text{spin}-1} = \mathbb{C}^2 \otimes \mathbb{C}^3$

$$\left\{ \begin{aligned} [\mathcal{H}, \text{SO}(3)] = 0 &\Rightarrow \mathcal{H} = \mathcal{H}_{2 \times 2} \otimes I_{3 \times 3} \\ U(g) &= I_{2 \times 2} \otimes U_\lambda(g) \end{aligned} \right\}$$

• Antiunitary symmetries and 0-fold way.

(1) time reversal

$$\left\{ \begin{aligned} \hat{T} \hat{\psi}_A \hat{T}^{-1} &= \sum_B (U_T^\dagger)_{A,B} \hat{\psi}_B \\ \hat{T} \hat{\psi}_A^\dagger \hat{T}^{-1} &= \sum_B \hat{\psi}_B^\dagger (U_T)_{B,A} \\ \hat{T} i \hat{T}^{-1} &= -i \end{aligned} \right. \quad \mathcal{T} = U_T \cdot K$$

$$\hat{T} \hat{H} \hat{T}^{-1} = \hat{H} \Leftrightarrow U_T \mathcal{H}^* U_T^\dagger = \mathcal{H}$$

$$\hat{T}^2 = \pm 1 \Leftrightarrow U_T U_T^* = \pm 1$$

$$T = \begin{cases} 0, & \text{if no } \mathcal{T} \text{ symm.} \\ +1, & \text{if } \mathcal{T}^2 = +1 \\ -1, & \text{if } \mathcal{T}^2 = -1 \end{cases} \rightarrow 3$$

(2) charge conjugation (particle-hole) symmetry.

$$\left\{ \begin{aligned} \hat{C} \hat{\psi}_A \hat{C}^{-1} &= \sum_B (U_C^*)_{A,B} \hat{\psi}_B^\dagger \\ \hat{C} \hat{\psi}_A^\dagger \hat{C}^{-1} &= \sum_B \hat{\psi}_B (U_C)_{B,A} \\ \hat{C} i \hat{C}^{-1} &= i \end{aligned} \right.$$

comes from T (transpose)

$$\hat{C} \hat{H} \hat{C}^{-1} = \hat{H} \Leftrightarrow U_C \mathcal{H}^* U_C^\dagger = -\mathcal{H}$$

$$C = \begin{cases} 0, & \text{if no } \mathcal{C} \text{ symm.} \\ +1, & \text{if } \mathcal{C}^2 = +1 \\ -1, & \text{if } \mathcal{C}^2 = -1 \end{cases} \rightarrow 3$$

(3) chiral (sublattice) symmetry.

$$\hat{S} = \hat{\gamma} \cdot \hat{C}$$

$$S = U_S = U_T \cdot U_C^*$$

$$\hat{S} \hat{H} \hat{S}^{-1} = \hat{H} \Leftrightarrow U_S \mathcal{H} U_S^\dagger = - \mathcal{H}$$

$$S = \begin{cases} 0, & \text{if no } S \text{ symm.} \\ 1, & \text{if } S \text{ symm.} \end{cases}$$

$$\text{if } T=C=0, \quad S=T \cdot C = \begin{cases} 0 \\ 1 \end{cases}$$

$$T=0, \pm \quad C=0, \pm \quad \begin{cases} T=C=0 \\ S=1 \end{cases}$$

In total, $3 \times 3 + 1 = 10$ classes

TABLE - “Ten Fold Way” [‘CARTAN Classes’]						Examples	
Name (Cartan)	T	C	S= TC	Time evolution operator $U(t) = \exp\{itH\}$	Anderson Localization NLSM Manifold G/H [compact (fermionic) sector]	SU(2) spin conserved	Some Examples of Systems
A (unitary)	0	0	0	U(N)	U(2n)/U(n)xU(n)	yes/no	IQHE Anderson
AI (orthogonal)	+1	0	0	U(N)/O(N)	Sp(4n) /Sp(2n)xSp(2n)	yes	Anderson
AII (symplectic)	-1	0	0	U(2N)/Sp(2N)	SO(2n)/SO(n)xSO(n)	no	Quantum spin Hall Z2-Top.Ins. Anderson(spinorbit)
AIII (chiral unitary)	0	0	1	U(N+M)/U(N)xU(M)	U(n)	yes/no	Random Flux Gade SC
BDI (chiral orth.)	+1	+1	1	SO(N+M)/SO(N)xSO(M)	U(2n)/Sp(2n)	yes/no	Bipartite Hopping Gade
CII (chiral sympl.)	-1	-1	1	Sp(2N+2M) /Sp(2N)xSp(2M)	U(n)/O(n)	no	Bipartite Hopping Gade
D	0	+1	0	O(N)	O(2n)/U(n)	no	(px+ipy)-wave 2D SC w/spin-orbit TQHE
C	0	-1	0	Sp(2N)	Sp(2n)/U(n)	yes	Singlet SC +mag.field (d+id)-wave SQHE
DIII	-1	+1	1	O(2N)/U(N)	O(n)	no	SC w/ spin-orbit He-3 B
CI	+1	-1	1	Sp(2N)/U(N)	Sp(2n)	yes	Singlet SC

(Ludwig 2015)

7.3. Examples of π I/TSC classification

2 complex classes $\begin{cases} A : TCS = 000 \\ AIII : TCS = 001 \end{cases}$

8 real classes $\begin{cases} \begin{cases} AI : TCS = +00 \\ AII : -00 \end{cases} \left. \vphantom{\begin{matrix} AI \\ AII \end{matrix}} \right\} \begin{matrix} \text{insulators} \\ G_f = U(1)_f \end{matrix} \\ \begin{cases} D : 0+0 \\ BDI : ++1 \\ DIII : -+1 \end{cases} \left. \vphantom{\begin{matrix} D \\ BDI \\ DIII \end{matrix}} \right\} \begin{matrix} \text{Superconductors} \\ G_f = \mathbb{Z}_2^f \end{matrix} \\ \begin{cases} C : 0-0 \\ CI : +-1 \\ CII : --1 \end{cases} \left. \vphantom{\begin{matrix} C \\ CI \\ CII \end{matrix}} \right\} \text{Superconductors with } SU(2)_s \text{ Symm.} \end{cases}$

• classification of 2D class A TI.

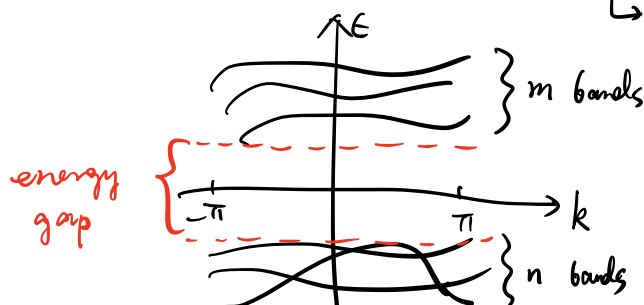
$TCS = 000$
 $\swarrow \searrow$
 γ broken insulator $\Rightarrow$ Chern insulator $\mathbb{Z}$

$\hat{H} = \sum_{A,B} \hat{\psi}_A^\dagger H_{AB} \hat{\psi}_B$
 $\downarrow$
 hermitian matrix $H^\dagger = H \Leftrightarrow H \in u(N) \Leftrightarrow e^{iH} \in U(N)$

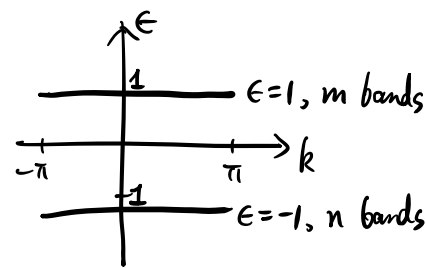
$\{\text{class A Hamiltonians}\} = u(N)$

If we have translational symmetry,

$\hat{H} = \sum_{\mathbf{r} \in \mathbb{Z}^2} \sum_{i=1}^{m+n} \epsilon_{ik} \hat{\psi}_{i\mathbf{k}}^\dagger \hat{\psi}_{i\mathbf{k}}$
 $\downarrow$
 band index.



homotopy
 (smoothly deform)



$$H(k) = U(k) \begin{pmatrix} \epsilon_1(k) & & \\ & \ddots & \\ & & \epsilon_{m+n}(k) \end{pmatrix} U(k)^\dagger \longrightarrow \tilde{H}(k) = U(k) \begin{pmatrix} I_m & \\ & -I_n \end{pmatrix} U(k)^\dagger$$

wavefunction unchanged
(U same)

$$H \in u(m+n)$$

$$\text{simplified } \tilde{H} \in \frac{U(m+n)}{U(m) \times U(n)}$$

For $U = \begin{pmatrix} U_m & \\ & U_n \end{pmatrix}$ with $U_m \in U(m)$, $U_n \in U(n)$,

$$\tilde{H} = U \begin{pmatrix} I_m & \\ & -I_n \end{pmatrix} U^\dagger = \begin{pmatrix} U_m U_m^\dagger & \\ & -U_n U_n^\dagger \end{pmatrix} = \begin{pmatrix} I_m & \\ & -I_n \end{pmatrix}$$

$$\{\text{class A simplified Hamiltonians}\} = \frac{U(m+n)}{U(m) \times U(n)} = C_0$$

2D class A TI

$$\Leftrightarrow \tilde{H}: B\mathbb{Z} = T^2 \longrightarrow C_0$$

$$\vec{k} \mapsto \tilde{H}(\vec{k})$$

Classification of 2D class A strong TI

$$= (\tilde{H}: S^2 \rightarrow C_0)$$

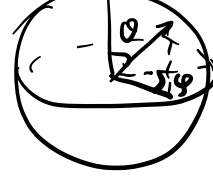
$$= \pi_2(C_0) = \mathbb{Z}$$

Consider the simpler case with $m=n=1$ (2 bands):

$$C_0 = \frac{U(2)}{U(1) \times U(1)} \cong S^2$$

$$U(2) \ni U = \begin{pmatrix} a & b \\ -e^{i\theta} b^* & e^{i\theta} a \end{pmatrix}, \quad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1$$

$$U \sim \begin{pmatrix} \cos \frac{\theta}{2} & e^{i\varphi} \sin \frac{\theta}{2} \\ -e^{-i\varphi} \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad \begin{matrix} 0 \leq \theta \leq \pi \\ 0 \leq \varphi < 2\pi \end{matrix}$$



$$\pi_2\left(\frac{U(2)}{U(1) \times U(1)}\right) = \pi_2(S^2) = \mathbb{Z}.$$

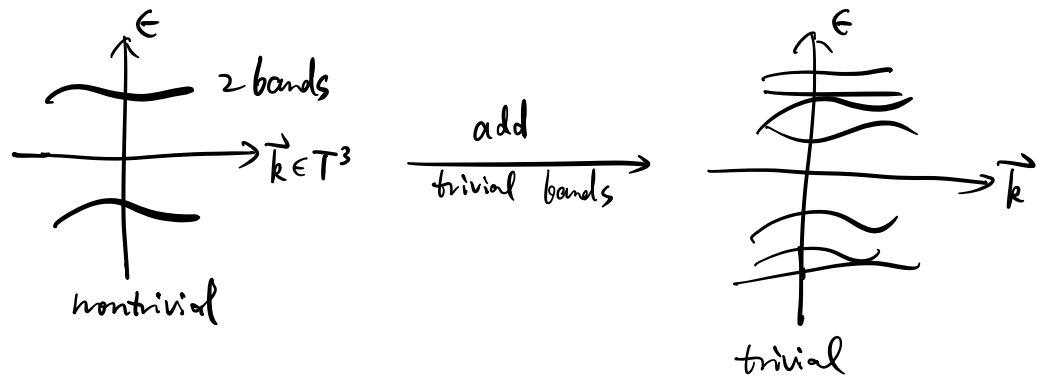
$$[T^2, S^2] = \mathbb{Z}$$

Remarks.

(1) Classification is obtained under stable condition

Example: 3D class A TI with 2 bands:

$$\begin{aligned} \tilde{H}: T^3 &\rightarrow \frac{U(2)}{U(1) \times U(1)} = S^2 \\ &\searrow \\ &S^3 \xrightarrow[\mathbb{Z}]{\text{Hopf}} \end{aligned}$$



Adding bands.

$$H = H \oplus H_0$$

$$H_0 = \begin{pmatrix} X & 0 \\ 0 & -X \end{pmatrix}, \quad X = U \begin{pmatrix} I_m & \\ & -I_n \end{pmatrix} U^\dagger$$

$$X^2 = I$$

$$\text{Claim. } H_0 \sim H_1 = \begin{pmatrix} & iI \\ -iI & \end{pmatrix}$$

$$\text{homotopy } H_t = \cos\left(\frac{\pi}{2}t\right) H_0 + \sin\left(\frac{\pi}{2}t\right) H_1$$

$$H_0^2 = (I \otimes X)^2 = I \otimes I = I$$

$$H_1^2 = (-\sigma_y \otimes I)^2 = I$$

$$H_0 H_1 = -H_1 H_0$$

$$\left. \begin{aligned} H_0^2 &= I \\ H_1^2 &= I \\ H_0 H_1 &= -H_1 H_0 \end{aligned} \right\} \Rightarrow H_t^2 = \left(\cos^2 \frac{\pi}{2} t + \sin^2 \frac{\pi}{2} t \right) I = I$$

$$\begin{pmatrix} X & \\ & -X \end{pmatrix} \underset{\text{gapped}}{\overset{\text{Symm.}}{\sim}} \begin{pmatrix} & iI \\ -iI & \end{pmatrix} \underset{\text{gapped}}{\overset{\text{Symm.}}{\sim}} \begin{pmatrix} I & \\ & -I \end{pmatrix}$$

$\{X\}_{\text{Symm.}}$ may have nontrivial topology,
but $\left\{\begin{pmatrix} X & \\ & -X \end{pmatrix}\right\}_{\text{Symm.}}$ is always contractible.

Equivalence of modes (with different number of bands)

• $H'_n \sim H''_n$ iff $H'_n \oplus Y_k \sim H''_n \oplus Y_k$ for some Y_k .

" $\Leftarrow$ ": $H' \oplus Y \sim H'' \oplus Y \Rightarrow H' \oplus Y \oplus (-Y) \sim H'' \oplus Y \oplus (-Y)$
 $\Rightarrow H' \sim H''$

• difference class $d(A, B) \rightarrow$ understood as $A \ominus B$

$(A'_n, B'_n) \sim (A''_m, B''_m)$ if $A'_n \oplus B''_m \sim A''_m \oplus B'_n$
 $\underbrace{A'_n \oplus B'_n}_{A' \ominus B'} \sim \underbrace{A''_m \oplus B''_m}_{A'' \ominus B''}$

This equi. gives a notion of equiv. of matrices with different sizes.

$(A_n, B_n) \sim (A_n \oplus (-B_n), B_n \oplus (-B_n)) = (A_n \oplus -B_n, \begin{pmatrix} I_n & \\ & -I_n \end{pmatrix})$

We can always choose $B_n = \begin{pmatrix} I_s & \\ & -I_s \end{pmatrix}$ with $n=2s$.

Invariant of (A, B) : $k = k(A) - k(B) = k(A) - s$
 $\downarrow$
 # negative energies of A

• consider again class A

For each k , space of $\tilde{H}$ of class A is $\frac{U(2s)}{U(s+k) \times U(s-k)}$

" $C_0 = \frac{U(m+n)}{U(m) \times U(n)}$ " $\rightarrow C_0 = \bigcup_{k \in \mathbb{Z}} \lim_{s \rightarrow \infty} \frac{U(2s)}{U(s+k) \times U(s-k)}$

(2) strong / weak TI / TSC.

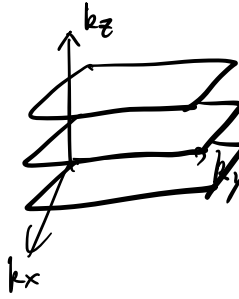
$B\mathbb{Z}^d = T^d \neq S^d$

$[T^d, C]_s \cong \pi_d(C) \oplus \bigoplus_{i=0}^{d-1} \binom{d}{i} \pi_i(C)$

Strong TI

Weak TI

stacking of strong TI of lower dims.



$$H_{3D}(k_x, k_y, k_z) = H_{2D}(k_x, k_y)$$

(3) We can use topological invariants to distinguish TI/TSC classes for each Symm class and dim.

(4) There are nontrivial (gapless, symmetric) edge states for nontrivial TI/TSC.

(5) Space of Hamiltonians for each Symm class:

Symmetric
No gapped condition

{ ① gapped
② flatten $\epsilon = \pm 1$ } $\Leftrightarrow \tilde{H}^2 = 1$

class	TCS	{ Hamiltonian H }	{ simplified Hamiltonian $\tilde{H}$ }
(Complex)	A	$u(N)$	$U(m+n)/U(m) \times U(n) = C_0$
	AIII	$u(m+N)/u(m) \times u(N)$	$u(N) = C_1$
(real)	AI	$u(N)/o(N)$	$O(m+n)/O(m) \times O(n) = R_0$
	BDI	$o(m+N)/o(m) \times d(N)$	$O(n) = R_1$
	D	$o(N)$	$O(2n)/U(n) = R_2$
	DIII	$so(2N)/u(N)$	$U(2n)/Sp(n) = R_3$
	AII	$u(2N)/sp(N)$	$Sp(m+n)/Sp(m) \times Sp(n) = R_4$
	CII	$sp(m+N)/sp(m) \times sp(N)$	$Sp(n) = R_5$
	C	$sp(N)$	$Sp(n)/U(n) = R_6$
	CI	$sp(N)/u(N)$	$U(n)/O(n) = R_7$

d -dim Strong TI/TSC are classified by $\pi_d(C_i \text{ or } R_i)$.